sparse to attempt a similar analysis in terms of training FLOPs, but we expect such an analysis may be more helpful in demonstrating a crisper relationship between training compute and downstream accuracy. When we accumulate the performance of models in Figure 29, so as to understand if scale appears more necessary to improve accuracy, we do see some notable jumps for many scenarios both in the range of 12B parameters and 50B parameters. These jumps are specifically attributable to T0++ (11B) and Anthropic-LM v4-s3 (52B), which likely instead indicates that attributing these jumps to scale is confounded by other changes (i.e. the training procedure of T0++ (11B) to deliberately improve few-shot prompting performance and the RLHF involved in Anthropic-LM v4-s3 (52B)). And we similarly see the relationship between perplexity (or, more accurately, log bits-per-byte) on **The Pile** and downstream accuracy is similarly messy (Figure 30), though we expect the confound of contamination (some models are trained on **The Pile** and others are not) exacerbates the messiness of these findings.

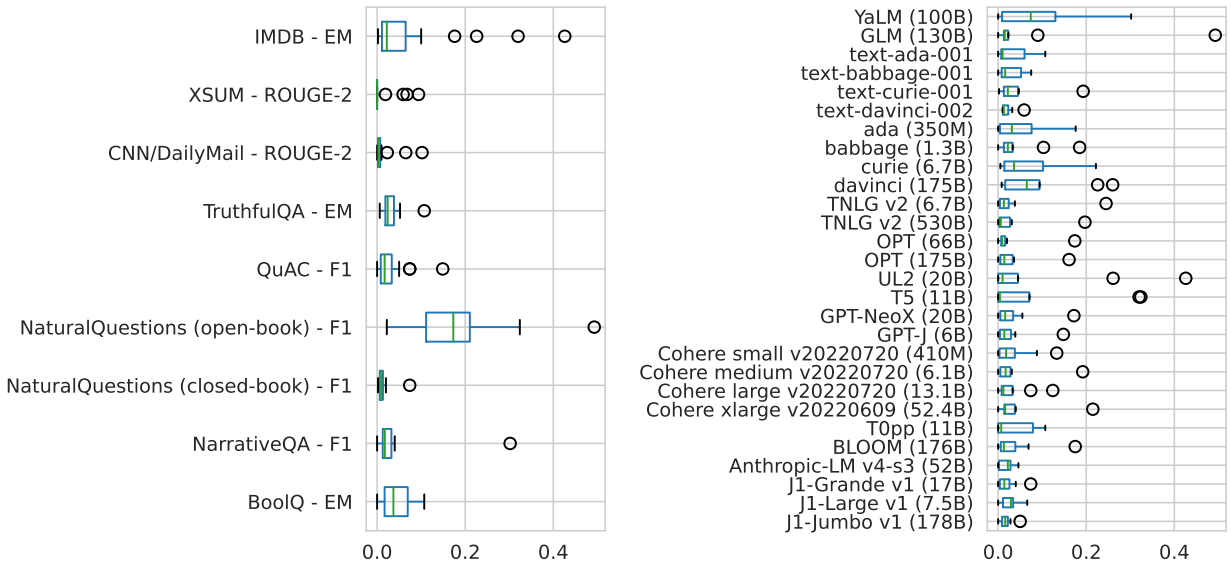

Figure 31: **Variance across seeds.** For a subset of models and scenarios, we evaluate each scenario with three different random sets of in-context examples. We compute the range of the accuracy metric (maximum minus minimum value over the three random seeds) and visualize across models and scenarios.

8.2 Prompting analysis

While the benchmark we design is general, we evaluate 30 models by adapting them through few-shot prompting (see §7: PROMPTING). Consequently, here we study variations across several of these models with respect to different design decisions involved in prompting.[66] Together, these analyses help clarify the potential brittleness of model behavior to prompt design, which bears on the legitimacy and reliability of using prompting to build practical systems.

Choice of in-context examples. Given that we adapt language models through few-shot prompting, the selection of the specific in-context examples could significantly influence model performance. In particular, unlike prior work (e.g. Brown et al., 2020), we use the same examples across all evaluation instances, rather than selecting different in-context examples for each evaluation instance, as this better reflects the realities of few-shot adaptation (Perez et al., 2021). (We describe our specific selection algorithm which, to first-approximation, is random selection from the scenario's training set while ensuring coverage of different classes for classification problems, in §J.2: PROMPTING-REMAINDER.) As a consequence, this exacerbates the variation observed for different choices of in-context examples (consistent with the high variability one might witness in practice if they truly only have a few examples).

To measure this sensitivity, we repeat all evaluations for 3 runs, meaning 3 choices of the in-context examples. To explore the sensitivity of model results to this choice, hover over a specific entry to see the variance across runs on the website. Further, the predictions across the 3 runs can be found for a given evaluation instance. Additionally, we visualize the accuracy range (best minus worst performance) across seeds in Figure 31. We observe that the performance of most models tends to be relatively consistent: the median range (across scenarios) is below 0.03 for 24 out of 28 models (exceptions are YaLM (100B), davinci (175B), curie (6.7B), ada (350M)). Interestingly, we find that there are scenarios where all models tend to exhibit higher variance. A prominent example is **NaturalQuestions** (open-book), where the median range is 0.1730 (e.g., davinci (175B) obtains F1 scores of 0.376, 0.611, and 0.636 across the three sets of in-context examples).

[66] We do not conduct all of these analyses on every model due to cost. We exclude all commercial models due to monetary cost, as well as TNLGv2 models due to long wait times given the scale of the model.

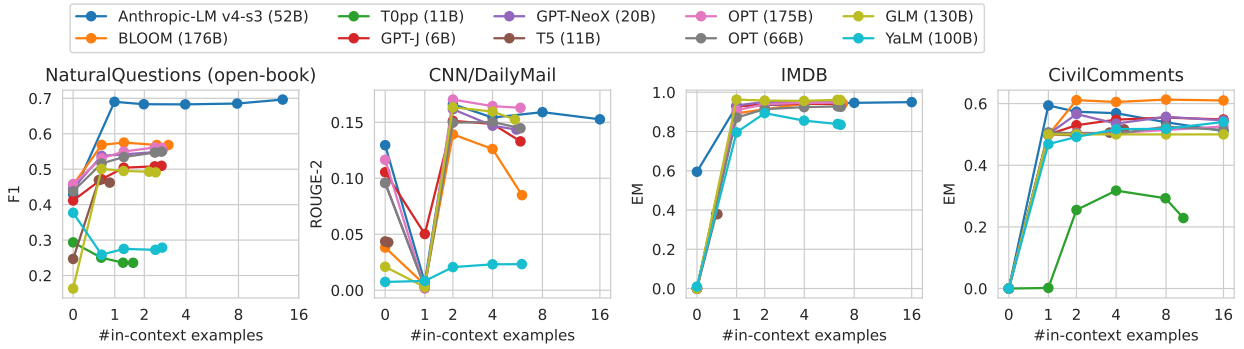

Figure 32: **Number of in-context examples.** For each model, we set the maximum number of in-context examples to [0, 1, 2, 4, 8, 16] and fit as many in-context examples as possible within the context window. We plot performance as a function of the average number of in-context examples actually used.

Number of in-context examples. By default, we either use 5 in-context examples, or fewer examples for scenarios where 5 examples do not fit within in the context window. To test how the number of examples (i.e. the sample efficiency of adaptation) influences performance, we vary the maximum number of examples across $n \in \{0, 1, 2, 4, 8, 16\}$. In Figure 32, we plot model performance as a fraction of the average number of in-context examples provided (which may be fewer than the maximum stated above if they do not fit inside the context window). To explore the results further, including the model generations, see `https://crfm.stanford.edu/helm/v0.1.0/?group=ablation_in_context`.

We find that all models show clear improvement from $n = 0$ to $n = 1$, sometimes having 0% accuracy in the zero-shot setting, with the consistent exception of **CNN/DailyMail** where zero-shot accuracy is better for almost all models. We posit that models may not effectively understand the appropriate length distribution and the poor reference summaries may comparatively mislead the model in the one-shot setting compared to the zero-shot setting. However, for larger numbers of in-context examples, we do not see consistent benefits across all models and all scenarios. The sole exception is OPT (175B) which, besides **CNN/DailyMail**, shows a perfectly monotonically increasing relationship between number of shots and model accuracy for **NaturalQuestions** (open-book), **IMDB**, and **CivilComments**.

Formatting of prompt. As we describe in §7: PROMPTING, beyond the in-context examples and the evaluation instance, there are several other details required to fully specify a prompt (e.g. instructions that describe what the model should do). Since this formatting exists in the space of natural language, it is difficult to specify concrete axes to systematically vary across (i.e. in contrast to how we can specify a range we consider for the number of in-context examples). Consequently, we consider the following motivated but fairly *ad hoc*/arbitrary changes to the prompt format involving instructions, input prefixes, output prefixes, and input suffixes. The exact changes to the prompt can be found at `https://crfm.stanford.edu/helm/v0.1.0/?group=ablation_prompts`, along with the results.

The clear finding is that the best prompt formatting is not consistent across models (i.e. models can stand to improve in their interoperability). In particular, one variants lead to an accuracy of 67.3% for Anthropic-LM v4-s3 (52B) on **NaturalQuestions** (open-book), whereas the prompt performs very poorly for BLOOM (176B), which drops from an accuracy around 60% to 8.5%. In some cases, we believe the prompting changes may lead to poor/undesirable interactions with the tokenizer, given the models use different tokenizers in several cases. For GLM (130B), we are intrigued to see the prompt involving mentioning the model is an expert AI assistant performs best across all four scenarios (**NaturalQuestions** (open-book), **CNN/DailyMail**, **IMDB**, and **CivilComments**) we look in terms of accuracy. Specifically, the prompt includes `instructions` of "I am an expert AI assistant who is here to help you with the following.", along with an `input_prefix` of "Passage: ", `input_suffix` of " "", and an `output_prefix` of "Answer:"